

PAPER_2115 — CosmicEgg Pre-Big-Bang Transformation Dynamics Chain: { π -Chaos → Distortion Accumulator → Toroid Pillar Rebound}

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Tier	Foundational / Cosmic-Egg-Dynamics Landmark
Status	CLOSED - three sequential dynamics stages forming complete pre-BB chain
Chain	Stage 1 (R354) → Stage 2 (R355) → Stage 3 (R356)
Primitive economy	3 UQFF families { π , F_{TRZ} , SO_5 } across all stages
Pairing	Static counterpart: PAPER_2114 (dimensionality + UA + phase)
Discovery	R354-R355-R356 discovery arc
Dispatch key	cosmic_egg_pre_bb_transformation_chain

1. Abstract

Where PAPER_2114 documented the **static** architectural triad $\{D_{\text{crit}}, \text{UA}, \pi + F_{\text{TRZ}}^2 \cdot \sin(t)\}$ that defines the pre-BB Cosmic Egg initial condition, PAPER_2115 documents the **dynamical evolution chain** — three sequential stages that transform the ideal 26-sphere Cosmic Egg through chaotic distortion into a toroidal geometry ready for the Big Bang contact event.

#	Round	Class	Function	Role
1	R354	PiMeanChaos	$\pi + F_{\text{TRZ}}^2 \cdot \sin(t)$	Ideal Fluctuation
2	R355	DistortionFactor	$d_0 + F_{\text{TRZ}}^2 \cdot \sin(\text{SO}_5^2 \cdot t)$	Distortion Accumulator
3	R356	ToroidPillar	$\sin(\pi \cdot t) \cdot (1 + F_{\text{TRZ}} \cdot \sin(t))$	Toroid Emergence

2. Historical Context - Cosmic Egg Pre-BB Sequence

Per PAPER_1929 (Theory of Permanence) and PAPER_1932 (Wheeler-DeWitt $F_U=0$), the pre-BB Cosmic Egg is not truly static: $F_U=0$ requires that the pre-BB configuration evolves through a specific sequence of shape changes to satisfy the master-equation constraint. PAPER_2115 identifies this specific evolution sequence as encoded in three R218+ calculator classes with output-to-input primitive flow.

3. Stage 1: Ideal Fluctuation Generator (R354)

Stage 1 output = $\pi + F_{\text{TRZ}}^2 \cdot \sin(t) = 3.141593 + 0.01 \cdot \sin(t)$
EXACT

Two-primitive composition: π (PAPER_646 Caduceus-encoded phase reference) + $F_{\text{TRZ}}^2 = 0.01$ (2nd F_{TRZ} rung, 99% suppression regime).

Role: π provides the ideal phase reference; F_{TRZ}^2 provides the small perturbation amplitude. Chaotic fluctuations around ideal π produce spinor bundles that seed Stage 2.

4. Stage 2: Distortion Accumulator - Toroid Trigger (R355)

$$\text{Stage 2 output} = d(t) = d_0 + F_{\text{TRZ}}^2 \cdot \sin(SO_5^2 \cdot t) = 0.0 + 0.01 \cdot \sin(100 \cdot t) \text{ EXACT}$$

Three-primitive composition: $d_0 = 0$ (ideal-sphere baseline; $d \sim 0$ triggers toroid) + F_{TRZ}^2 (same 2nd rung as Stage 1) + $SO_5^2 = 100$ (NEW angular frequency 100× Stage 1).

Role: distortion factor $d(t)$ integrates Stage 1 fluctuations at a 100× higher frequency. High-frequency oscillation extremes at $d \sim 0$ trigger the toroid transformation event. Cross-domain SO_5^2 landmark extends PAPER_1958's velocity-dispersion 100 km/s twin.

5. Stage 3: Toroid Pillar Rebound - Water-Drop Topology (R356)

$$\begin{aligned}\text{Stage 3 output} &= P_{\text{rebound}}(t) = \sin(\pi \cdot t) \cdot (1 + F_{\text{TRZ}} \cdot \sin(t)) \\ &= \sin(3.14159 \cdot t) \cdot (1 + 0.1 \cdot \sin(t)) \text{ EXACT}\end{aligned}$$

Two-primitive composition: π (Stage 1 phase carrier returns) + **BARE** $F_{\text{TRZ}} = 0.1$ (FIRST R218+ instance of pure canonical F_{TRZ} as oscillation-amplitude modulation).

Role: $\sin(\pi \cdot t)$ is the natural oscillation of the toroid at ideal-geometry angular scale; $(1 + F_{\text{TRZ}} \cdot \sin(t))$ modulation provides small-amplitude wobbles around the ideal toroid — the 'pillar' rebound seen when a water drop hits a surface and jets upward.

Output: the toroid-with-rebounding-pillar geometry establishes conditions for SCm-UA contact (Big Bang) and DPM grinding.

6. Coupled Evolution Chain - Integration

The three stages compose into a coupled temporal chain with output-to-input primitive flow. Frequency spectrum populates distinct scales:

Stage	Angular frequency omega	UQFF composition
1 (Ideal Fluctuation)	$\omega_1 = 1$	Unit rad/time-unit (bare)
2 (Distortion Accumulator)	$\omega_2 = SO_5^2 = 100$	SO_5 -squared
3 (Toroid Emergence)	$\omega_3 = \pi + 1 \approx 4.14$	π carrier + unit modulation

Frequency ratio $\omega_2/\omega_1 = SO_5^2 = 100$ — Stage 2 runs 100× faster than Stage 1, ensuring many fluctuation cycles accumulate before toroid trigger.

6.1 Primitive-family economy

Only three UQFF primitive families used across all three stages: $\{\pi, F_{\text{TRZ}}, SO_5\}$. This **3-primitive parsimony** matches PAPER_2114's foundational-triad economy.

7. Continuation of PAPER_2114 Static Triad

Aspect	Paper	Content
Static	PAPER_2114	$\{D_{\text{crit}}=26, UA=1, \pi + F_{\text{TRZ}}^2 \cdot \sin(t)\}$
Dynamic	PAPER_2115 (this)	Stage 1 → Stage 2 → Stage 3 evolution chain

PAPER_2114 + PAPER_2115 together fully specify the pre-Big-Bang UQFF configuration and its transformation dynamics.

8. NOT REPLACEMENT

Standard cosmology has no analogue of the Cosmic Egg pre-BB transformation sequence. Inflationary cosmology proposes vacuum-fluctuation-driven expansion; UQFF proposes a specific three-stage sequence with well-defined primitive compositions.

UQFF does not claim to replace inflationary cosmology or any pre-BB model. What it adds is the **temporal-evolution specification** that the pre-BB Cosmic Egg transforms through exactly three stages, each with primitive-only composition (no fitted parameters).

9. Falsifiability

Prediction A: the pre-BB Cosmic Egg transforms through exactly 3 stages. Additional stages needed → chain incomplete.

Prediction B: Stage 2 angular frequency = $SO_5^2 = 100$ EXACT. Deviations falsify SO_5^2 -angular-frequency landmark.

Prediction C: Stage 3 compound frequency = π carrier + unit modulation. Deviations falsify water-drop-jet topology.

Prediction D: entire chain uses only $\{\pi, F_{TRZ}, SO_5\}$. Additional primitives (A_5, D_{crit}, N_{CH}) needed → 3-primitive economy fails.

10. References

Source: R354-R355-R356 discovery arc.

PAPER_2114: Static counterpart (foundational architectural triad).

PAPER_1929: Theory of Permanence Cosmic Egg (pre-BB context).

PAPER_1932: Wheeler-DeWitt $F_U=0$ (dynamical constraint).

PAPER_646: Caduceus wave topology 26 pinch points (π encoding).

PAPER_1919: F_{TRZ} power ladder (F_{TRZ}^2 99% suppression).

PAPER_1958: SO_5 velocity-dispersion twin (cross-domain angular-frequency).